Figure 1 | Capability-resolved evaluation of PPI ΔΔG predictors

A Provenance-filtered benchmark construction

SKEMPI v2.0

experimentally measured PPI ΔΔG mutations

4,473 rows

214 targets

108 families



PPI target + mutation measurements

Predictions from diverse method families

Physics | Structure ML | Protein LM | Graph NN | ...

Provenance & leakage audit

- ✓ SHA-256 row-set fingerprints
- ✓ retain provenance-clean Family-OOD
- × remove training overlap / permissive splits

Primary matched evaluation universe

3,351 mutation rows

102 targets (n ≥ 5)

10 method families

B Exact covariance-signal decomposition

True ΔΔG

$$y_{ci} = \mu_c + \varepsilon_{ci}$$

μ_c: target mean • ε_{ci}: within-target deviation

Predicted ΔΔG

$$\hat{y}_{ci} = \hat{\mu}_c + \hat{\varepsilon}_{ci}$$

$\hat{\mu}_c$: predicted target mean • $\hat{\varepsilon}_{ci}$: predicted deviation

$$r_{\text{pooled}} = C_B + C_W$$

C_B

between-target signal

target-level scale placement



correlation of target means

C_W

within-target signal

mutation ordering within a target



correlation of within-target deviations

Exact identity: residual $\leq 3.3 \times 10^{-16}$

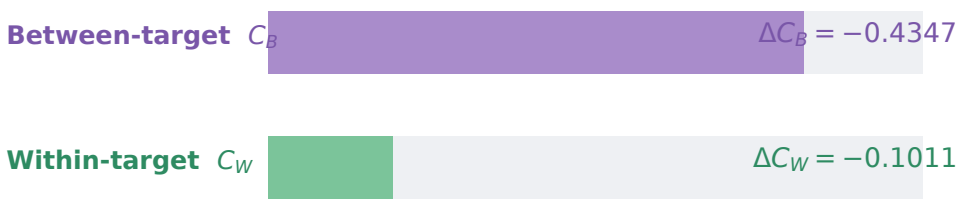
common-scale additive covariance contributions — not independent correlations

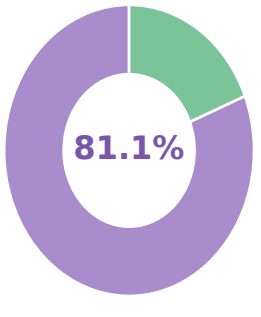
C Controlled split collapse on identical 7,523 rows

Same mutation rows and model configurations; only split strictness changes

Random	Complex	seq40	Family-OOD
(permutation)	(interface)	(sequence)	(between-family)
0.6844	**0.1931**	**0.1961**	**0.1508**
pooled Pearson	pooled Pearson	pooled Pearson	pooled Pearson

Decomposition of performance drop: Random → Family-OOD





81.1% of the summed absolute component change

is carried by loss of between-target signal.

C_W remains positive under Family-OOD (+0.1580).

D Retrospective ablation trajectory on the 3,351-row universe

Matched local-physics ablations; raw values from Supplementary Fig. S2

		Pooled Pearson	Within-target Macro Spearman ρ
1	Mutation-only baseline (F1)	**0.361**	**0.070**
2	+ Static local geometry (F2)	**0.389**	**0.068**
3	+ Mutation × partner perturbation (F4)	**0.431**	**0.113**
4	**Full local-physics model (F8)**	**0.466**	**0.145**

